

ISHA PATRO

isha.patro@stern.nyu.edu | +1 (347) 241-6603 | [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | New York City, 11213

EDUCATION

NEW YORK UNIVERSITY

Master of Science in **Financial Engineering**

GPA: 3.6/4.0

New York, NY

Expected 05/27

VELLORE INSTITUTE OF TECHNOLOGY, AP

Bachelor in **Computer Science**

GPA: 3.5/4.0

Andhra Pradesh, India

07/21

PROGRAMMING & TECHNICAL SKILLS

Programming & Development: Python, JavaScript, HTML, CSS, VBA, React, REST APIs, Streamlit, Dash, AG Grid, Tableau

Data Science: Statistical Modeling, Machine Learning, Deep Learning, Attention Mechanisms, Large Language Models (LLMs), Generative AI, RAG, Agentic AI, Model Context Protocol (MCP), Natural Language Processing (NLP), Fine-Tuning, AI Inference, LLM Evaluation, Prompt Engineering

Databases & Cloud: SQL, q/kdb+, AWS, MongoDB, Microsoft Office, Microsoft Excel, Bloomberg Terminal

Quantitative Finance: Fixed Income, Equity Research, Portfolio Management, Risk Management, Time Series Forecasting, Prediction Markets

EXPERIENCE

FINANCIAL ARTIFICIAL INTELLIGENCE WORKING GROUP

New York, NY

Quantitative Finance Research Intern

07/26 - 09/26

- Developed a corpus for Fixed Income Modelling, Credit Card Securitization, and Derivatives workflows to benchmark AI agent outputs against Excel-based models, improving calculation accuracy, reasoning traceability, and hallucination resistance.

J.P. MORGAN CHASE & CO. | Commercial & Investment Bank

Bengaluru, India

Software Developer Engineer II, MBS Division - Fixed Income Pricing Team

01/21 - 08/25

- Led the development and post-launch ownership of Trade Activity Monitoring (TAM) as Epic Owner, collaborating with stakeholders to achieve a 50% latency reduction and significantly improve trading floor execution efficiency.
- Pioneered and architected a comprehensive Mortgage Matrix modernization initiative driving 12% latency improvement supporting \$5B+ annual trading volume.
- Designed and implemented Trade Reporting and Compliance Engine (TRACE) in an agile environment collaborating with cross-functional teams to boost data processing efficiency by 30% and streamline regulatory reporting workflows.

AMAZON

Remote

ACMS Intern

05/20 - 07/20

- Maintained and enhanced a COVID-19 data API in Python, integrating government health data, zone thresholds from a partner team's API, and multiple shipping carrier feeds to compute delivery-time estimates, validated and normalized inputs, and applied caching and scheduled refreshes to deliver accurate, low-latency data to the downstream teams.

PUBLICATIONS / OPEN SOURCE CONTRIBUTION

- Patro, Isha (2026) "Modeling Volatility in Polymarket Prices with Binary Jump-Tweedie Dynamics," Under review for the Prediction Markets Conference, Yale School of Management & Swensen Asset Management Institute.
- Patro, Isha and Appana, Raj (2025) "FinOKF: Evidence-Aware Finance Open Knowledge Formatting with LRU Caching for Auditable Equity Research," Under review at Journal of Association for the Advancement of Artificial Intelligence.
- Patro, Isha (2026) "Implied Prepayment Surfaces via Finance-Informed Neural Networks in Agency MBS Markets," Under review at Journal of Computational Finance.
- Patro, Isha (2024) "[Unlocking Financial Wisdom: Building Knowledge Graphs from Investopedia](#)," *International Journal of Research and Analytical Reviews (IJRAR)*, Sep 2024.
- Patro, Isha (2025) "[qutePandas](#): A Pandas-Compatible Execution Layer Backed by kdb+/q, Delivering 10x–100x+ Speedups on Large-Scale Analytical Workloads," Open Source Contribution. Published on PyPI, June 2025.

HONORS / AWARDS

- Awarded the prestigious **Numerix Women in Finance Scholarship**, a **\$20,000 honor**, 2025
- Two-time 1st Place Winner, Engineering Clinics Expo, VIT (**\$1,200**) - across all engineering disciplines, 2017-2018

ACADEMIC PROJECTS

Spline-Based Conditional Market Beta & Statistical Arbitrage Framework

09/26

- Modelled a NumPy-based cubic spline regression and k-fold cross-validated truncated basis engine to capture state-dependent non-linear asset co-movements, extracting idiosyncratic residual spreads for market-neutral statistical arbitrage.

Nike Equity Research Dashboard | DCF Valuation & Competitive Channel Analysis

09/26

- Built a data-driven equity research platform integrating SEC filings, DCF valuation, peer benchmarking, sentiment-adjusted price targets, and wholesale-channel analysis across Deckers, On, and Foot Locker.

Money Talks

09/26

- Engineered an AI-powered SaaS analytics platform that surfaces revenue concentration risk, tracks high-value customer churn, and evaluates go-to-market performance alongside underlying monthly recurring revenue (MRR) health.

- 2WIN** 06/26
- Designed and backtested quantitative prediction-market strategies using tick-level order-book data, dynamic risk-based sizing, automated execution logic, and portfolio PnL attribution across real-time Polymarket markets.
- VORTEX** 05/26
- Architected a full-stack quantitative risk analytics platform: Gaussian HMM regime detection, 5-method VaR backtesting, XGBoost volatility forecasting, and a LangGraph multi-agent LLM system.
- Deep Temporal Models With Attention and Skip Connections** 03/26
- Architected an end-to-end commodity intelligence pipeline combining FRED macroeconomic data, FinBERT news embeddings, dynamic graph learning, hybrid hyperbolic attention, and deep temporal models for futures-direction forecasting.
- Geometry of Intelligence** 01/25 - Present
- Created an independent machine learning research series explaining algorithms from first principles through mathematical intuition, probability, and clean code implementations.
- Cross-Currency Dynamics in Cryptocurrencies under Stablecoin Regulation** 01/26 - 03/26
- Analyzed cross-currency basis and liquidity fragmentation in crypto markets using high-frequency data across exchanges, evaluating stablecoin-driven pricing inefficiencies and regulatory impact (GENIUS Act)
- Monte Carlo Pricing of European Barrier Options** 10/25
- Developed a Monte Carlo engine under risk-neutral GBM to price path-dependent barrier options (Down-and-Out, Knock-In/Out), incorporating barrier monitoring and sensitivity analysis.
- Fixed Income Analytics & Securitization Suite** 09/25
- Built a financial modeling suite in Excel featuring Yield Curve Calibration using Nelson-Siegel model, Mortgage-Backed Securities (MBS) delta-neutral hedging, and multi-tranche credit card ABS cash flow waterfalls.
- Advanced NASDAQ Market Prediction System with Deep Learning Optimization** 08/25
- Engineered a sophisticated LSTM neural network system for high-frequency NASDAQ market prediction, achieving 26.7% improvement in forecasting accuracy through advanced hyperparameter optimization.
- Minted Wealth** 03/25
- Developed a comprehensive financial intelligence dashboard integrating live market data, portfolio performance analytics, and net worth insights for smarter investment decisions.